

Lab 2

Calculating the Difference between Two Consecutive Images

CSI 4133 Computer Methods in Picture Processing and Analysis

Fall 2024

School of Electrical Engineering and Computer Science

University of Ottawa

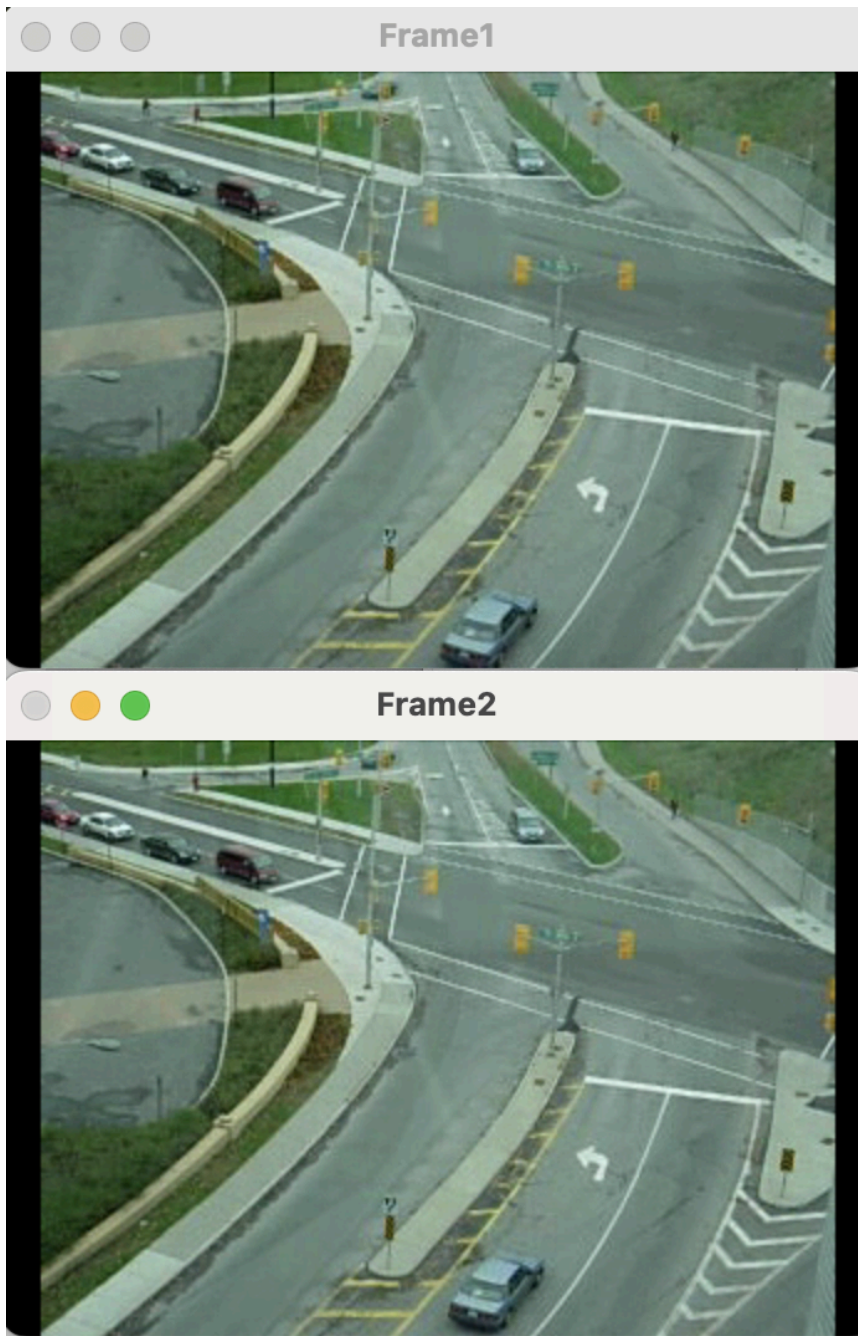
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Screenshots

- Original two frame chosen



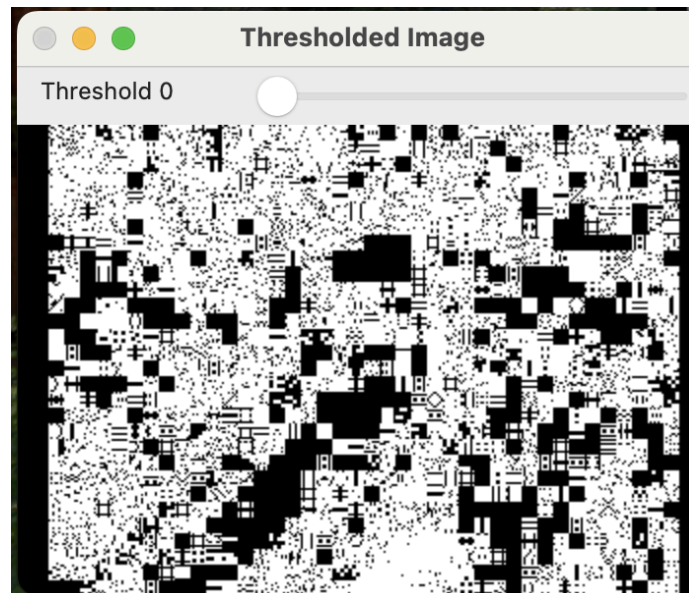
- Convert RGB image to GrayScale Image



- Windows with Threshold Track Bar



- Different Threshold value images
 - Threshold value = 0



- Threshold value = 15



- Threshold value = 30

The best result is achieved with a threshold value of 30, which highlights most of the car's pixels while minimizing background pixels, thereby reducing noise that could interfere with detection.



- Threshold value = 50



Source Code

```
# Lab2
# Name: Zijun Ye
# Student Number: 300168065
import cv2

#1 load image
carFrame1 = cv2.imread('images01/car1.bmp')
carFrame2 = cv2.imread('images01/car2.bmp')

#2 Convert RGB image to gray scale image
# use method cvtColor(src,bwsrc, cv::COLOR_RGB2GRAY)
grayCarFrame1 = cv2.cvtColor(carFrame1,cv2.COLOR_RGB2GRAY)
grayCarFrame2 = cv2.cvtColor(carFrame2,cv2.COLOR_RGB2GRAY)

cv2.imshow('grayFrame1',grayCarFrame1)
cv2.imshow('grayFrame2',grayCarFrame2)

#3 Calculate the pixel intensity difference between two images (absolute
value)
# can either be subtract OR absdiff(give absolute value)
frame_difference = cv2.absdiff(grayCarFrame1,grayCarFrame2)

#create a window to display
cv2.namedWindow('Thresholded Image')
```

```
#4 Perform thresholding on the difference image to get areas of movement in
binary form
# threshold(src input image, threshold value, new value value,max value,
thresholding type )
def update_threshold(val):
    thresh_value, thresholded = cv2.threshold(frame_difference, val , 255,
cv2.THRESH_BINARY)
    cv2.imshow('Thresholded Image', thresholded)

#5 create the track bar ("track bar name", "window name", initial value,
max value, call back function)
cv2.createTrackbar('Threshold', 'Thresholded Image', 30, 255,
update_threshold)

update_threshold(30)

# Wait for a key press and close the window
cv2.waitKey(0)
cv2.destroyAllWindows()
```